

EXERCISE SOLUTIONS TO *THE RISING SEA: FOUNDATIONS OF ALGEBRAIC GEOMETRY*

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1. Just enough category theory to be dangerous

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1. JUST ENOUGH CATEGORY THEORY TO BE DANGEROUS

Exercise (1.1.C.). Fix a field k . Let $(-)^{\vee\vee}$ be the double dual functor from the category of finite-dimensional k -vector spaces to itself. Show that $(-)^{\vee\vee}$ is naturally isomorphic to the identity functor.

Proof. For a fixed vector space V , define the map $\alpha_V : V \rightarrow V^{\vee\vee}$ by $v \mapsto (\phi \mapsto \phi(v))$. This is injective because if $\phi \mapsto \phi(v)$ is the zero map for every ϕ , then $v = 0$, so by finite-dimensionality α_V is an isomorphism. And it remains to check that for any map of vector spaces $f : V \rightarrow W$ and $v \in V$, that

$$\begin{aligned}\alpha_W(f(v)) &= (\phi \mapsto \phi(f(v))) \\ &= (\psi \mapsto \psi(v)) \circ f^\vee \\ &= f^{\vee\vee}(\psi \mapsto \psi(v)) \\ &= f^{\vee\vee}(\alpha_V(v))\end{aligned}$$

which completes the proof. □

Exercise (1.1.D). Let $\mathcal{V}$ be the category whose objects are the k -vector spaces k^n for $n \geq 0$, and whose morphisms are linear transformations. There is a functor F from $\mathcal{V}$ to the category of f.d. k -vector spaces sending each vector space to itself. Show that F is an equivalence of categories.

Proof. Following Vakil's remark about ignoring set-theoretic conditions, choose a basis for each finite dimensional k -vector space. For any vector space of the form k^n , choose the standard basis $\{e_i\}$.

Then define a functor G from f.d. k -vector spaces to $\mathcal{V}$ by sending an n -dimensional k -vector space to k^n . For a morphism ϕ from V to an m -dimensional vector space W , let $\{v_1, \dots, v_n\}$ be the chosen basis for V and $\{w_1, \dots, w_m\}$ the chosen basis for W . Then define $G\phi$ to be the matrix whose (i, j) th entry is the coefficient of w_j in $\phi(v_i)$.

Define a natural transformation α from FG to the identity functor where $\alpha_V : FG V \rightarrow V$ sends $FG V = k^n$ to V by mapping the standard basis vector e_i of $FG V$ to the chosen i th basis vector v_i of V . Then α_V is an isomorphism. For naturality, we have for a map $\phi : V \rightarrow W$ that

$$\begin{aligned}\phi(\alpha_V(e_i)) &= \phi(v_i) \\ &= \sum_j (G\phi)_{i,j} w_j \\ &= \alpha_W\left(\sum_j (G\phi)_{i,j} e_j\right) \\ &= \alpha_W(FG\phi)(e_i).\end{aligned}$$

In the other direction, GF is the identity functor, which completes the proof. □

Exercise (1.2.A). Show that any two initial objects are uniquely isomorphic, and any two final objects are uniquely isomorphic.

Proof. Let i_1, i_2 be initial. Then there is exactly one map $f_{12} : i_1 \rightarrow i_2$ and likewise $f_{21} : i_2 \rightarrow i_1$. Since there is also exactly one map from i_1 to itself, the identity, so because $f_{21} \circ f_{12}$ is one such map, it must be the identity. Likewise $f_{12} \circ f_{21}$ is the identity, which yields the unique isomorphisms. The argument is exactly the same for final objects. $\square$

Exercise (1.2.B). What are the initial and final objects (if they exist) in Ring, Set, and Top, or the category of subsets of a set under inclusion, or the category of open subsets of a topological space under inclusion?

Solution. In Ring, the initial object is the integers, since 1 mapping to 1 dictates where the remaining integers go; the terminal object is the zero ring, since everything must map to zero. In Set, the initial object is the empty set; the final object is the singleton. In Top, the initial object is the empty space; the final object is the singleton. In the category of subsets or open sets, the initial object is the empty set/space; the terminal object is the entire set/space. $\square$

Exercise (1.2.C). Let A be a commutative ring and S a multiplicative subset of A . Show that $f : A \mapsto S^{-1}A$ via $a \mapsto a/1$ is injective if and only if S contains no zerodivisors.

Proof. First suppose f is not injective, i.e. there is a nonzero element $a \in A$ such that $a/1 = 0/1$. So there exists some $s \in S$ such that $s(a - 0) = sa = 0$. Then s is a zerodivisor.

Conversely, suppose S contains a zerodivisor s , with $st = 0$ for t nonzero. Then $0 = st = s(t - 0)$, $t/1 = 0/1 = 0$, so $f(t) = t/1 = 0/1$ and f fails to be injective. $\square$

Exercise (1.2.D). Verify that $\iota : A \rightarrow S^{-1}A$ is initial among A -algebras B where every element of S is sent to an invertible element in B , i.e. any map $\phi : A \rightarrow B$ sending all elements of S to invertible elements in B factors through $S^{-1}A$.

Proof. Define $\tilde{\phi} : S^{-1}A \rightarrow B$ via $\tilde{\phi}(a/s) := \phi(a)\phi(s)^{-1}$. This makes sense since we have assumed that $\phi(s)$ is invertible. Let $a, b \in A$ and $s, t \in S$. Then

$$\tilde{\phi}(ab/st) = \phi(ab)\phi(st)^{-1} = \phi(a)\phi(b)\phi(s)^{-1}\phi(t)^{-1} = \tilde{\phi}(a/s)\tilde{\phi}(b/t)$$

and

$$\begin{aligned} \tilde{\phi}(a/s + b/t) &= \tilde{\phi}((at + sb)/st) \\ &= \phi(at + sb)\phi(st)^{-1} \\ &= \phi(a)\phi(t)/\phi(s)\phi(t) + \phi(s)\phi(b)/\phi(s)\phi(t) \\ &= \tilde{\phi}(a/s) + \tilde{\phi}(b/t). \end{aligned}$$

so $\tilde{\phi}$ is a ring homomorphism. And $\tilde{\phi}(\iota(a)) = \tilde{\phi}(a/1) = \phi(a)/\phi(1) = \phi(a)$ as desired. $\square$

Exercise (1.2.E). The localization of an A -module M defined via the universal property, actually exists.

Proof. Following the hint, define $S^{-1}M$ to have elements of the form m/s where $m \in M$ and $s \in S$, subject to the equivalence relation where $m/s \sim m'/t$ if there exists an element $s' \in S$ such that $s'(ms - m't) = 0$, with the standard operation and $S^{-1}A$ -module operation. Define $\iota : M \rightarrow S^{-1}M$ via $m \mapsto m/1$.

Let $\phi : M \rightarrow N$ be an A -module map such that multiplication by any element of s is an isomorphism $N \rightarrow N$. Then we can define a map $\tilde{\phi} : S^{-1}M \rightarrow N$ via $m/s \mapsto s^{-1}\phi(m)$, which makes sense taking s^{-1} to be the inverse to multiplication by s . This is an A -module map and $\tilde{\phi}(\iota(m)) = \tilde{\phi}(m/1) = 1^{-1}\phi(m) = \phi(m)$. $\square$

Exercise (1.2.F). *Properties of localization.*

- (1) *Localization commutes with finite products, i.e. for A -modules $M_1, \dots, M_n$ provide an A -module isomorphism $S^{-1}(M_1 \times \dots \times M_n) \rightarrow S^{-1}M_1 \times \dots \times S^{-1}M_n$.*

Proof. Define our map via $(m_1, \dots, m_n)/s \mapsto (m_1/s, \dots, m_n/s)$. For surjectivity, given an element $(m_1/s_1, \dots, m_n/s_n)$, the element

$$(s_2 \cdots s_n m_1, \dots, s_1 \cdots s_{n-1} m_n) / s_1 \cdots s_n$$

will map to it. For injectivity, suppose $(m_1/s, \dots, m_n/s)$ is zero, i.e. there exist elements $t_i \in S$ such that $t_i m_i = 0$ for all i . Then $t_1 \cdots t_n (m_1, \dots, m_n) = 0$, so $(m_1, \dots, m_n)/s = 0$. $\square$

- (2) *Localization commutes with arbitrary direct sums.*

Proof. Let Λ be an indexing set. Define a map $S^{-1}(\bigoplus_{\lambda \in \Lambda} M_\lambda) \rightarrow \bigoplus_{\lambda \in \Lambda} S^{-1}M_\lambda$ via

$$(m_\lambda)_{\lambda \in \Lambda} / s \mapsto (m_\lambda / s)_{\lambda \in \Lambda}.$$

For surjectivity, given an element $(m_\lambda / s)_\lambda$, we know that only finitely many indices are nonzero, say $\lambda_1, \dots, \lambda_k$. Then let $s = s_{\lambda_1} \cdots s_{\lambda_k}$, and consider the element $(k_\lambda m_\lambda)_{\lambda \in \Lambda}$ where $k_{\lambda_i} = s/s_{\lambda_i}$ for the nonzero indices and zero otherwise. Then it maps to what we want. For injectivity, suppose $(m_\lambda / s)_{\lambda \in \Lambda}$ is zero, i.e. there exist elements t_λ such that $t_\lambda m_\lambda = 0$ for all λ . Again only finitely many of the m_λ are nonzero, so take the corresponding t_λ and let their product be t . Then $t(m_\lambda)_{\lambda \in \Lambda} = 0$ and we are done. $\square$

- (3) *Localization does not commute with infinite products, i.e. the same map defined as above is not always an isomorphism for infinite products.*

Proof. We know that $\mathbf{Z}$ localized at $\mathbf{Z} \setminus \{0\}$ is $\mathbf{Q}$, so that $\prod_{\mathbf{N}} \mathbf{Q}$ contains for example the element $x := (1, 1/2, 1/3, \dots)$. On the other hand, suppose there were an element in $(\mathbf{Z} \setminus \{0\})^{-1}(\mathbf{Z}^{\mathbf{N}})$ mapping to x , i.e. some element $(a_1, a_2, \dots)/s$ where $a_i \in \mathbf{Z}$ and $s \in \mathbf{Z} \setminus \{0\}$ such that $a_n/s = 1/n$ for all $n \in \mathbf{N}$. Then $na_n = s$ for all $n \in \mathbf{N}$, i.e. s is an integer divisible by all natural numbers, a contradiction. $\square$

Exercise (1.2.G). *Show that $\mathbf{Z}/10 \otimes \mathbf{Z}/12 \cong \mathbf{Z}/2$.*

Proof. We define a map $\tilde{f} : \mathbf{Z}/10 \times \mathbf{Z}/12 \rightarrow \mathbf{Z}/2$ via $([a], [b]) \mapsto [ab]$. This is clearly bilinear. If $a = a' + 10n$ and $b = b' + 12m$, so $[ab] = [a'b' + 12ma' + 10nb' + 120nm] = [a'b']$ in $\mathbf{Z}/2$, so this map is well defined. By the universal property of tensor products, $\tilde{f}$ factors through the tensor product to yield a map $f : \mathbf{Z}/10 \otimes \mathbf{Z}/12 \rightarrow \mathbf{Z}/2$ such that $[a] \otimes [b] \mapsto [ab]$.

Consider the map $g : \mathbf{Z}/2 \rightarrow \mathbf{Z}/10 \otimes \mathbf{Z}/12$ given by $[a] \mapsto [a] \otimes [1]$. This is again bilinear, and if $a = a' + 2n$, then $[a] \otimes [1] = [a' + 2n] \otimes [1] = [a'] \otimes [1] + [12n - 10n] \otimes [1] = [a'] \otimes [1] + [1] \otimes [12n] - [10n] \otimes [1] = [a']$, so this is well defined.

We moreover have $fg([a]) = f([a] \otimes [1]) = [a]$ and $gf([a], [b]) = g([ab]) = [ab] \otimes [1] = [a] \otimes [b]$. Hence f is an isomorphism. (Note that this argument works to show in general that $\mathbf{Z}/m \otimes \mathbf{Z}/n \cong \mathbf{Z}/\gcd(m, n)$.) $\square$

Exercise (1.2.H). *Show that $(-) \otimes_A N$ is a covariant functor $\text{Mod}_A \rightarrow \text{Mod}_A$, and that it is right exact.*

Proof. Let $f : M \rightarrow M'$ be an A -module map. Then $(m, n) \mapsto f(m) \otimes n$ is bilinear, so it yields a map $f_* : M \otimes N \rightarrow M' \otimes N$, which on generators is given by $m \otimes n \mapsto f(m) \otimes n$. This respects composition, thus making $(-) \otimes N$ into a functor.

Let $M \xrightarrow{f} M' \xrightarrow{g} M'' \xrightarrow{h} 0$ be exact. Tensoring, we get a complex

$$M \otimes N \xrightarrow{f_*} M' \otimes N \xrightarrow{g_*} M'' \otimes N \xrightarrow{h_*} 0,$$

which we need to check is exact. We know by the first isomorphism theorem that g induces an isomorphism $M'/\text{im}(f) \rightarrow M''$.

Claim 1.1. Tensor products commute with quotients.

Proof. Let P, Q, R be A -modules. Suppose Q is a submodule of P . Consider the map $\tilde{f} : (P/Q) \times R \rightarrow (P \otimes R)/(Q \otimes R)$ given by $([p], r) \mapsto [p \otimes r]$. If $[p] = [p']$, then $[p \otimes r] - [p' \otimes r] = [p - p' \otimes r] = [q \otimes r] = 0$ so this map is well defined. One can also check bilinearity. Hence it extends to a map $f : (P/Q) \otimes R \rightarrow (P \otimes R)/(Q \otimes R)$.

We now define a map $\tilde{g} : P \times R \rightarrow (P/Q) \otimes R$ via $(p, r) \mapsto [p] \otimes r$. This is bilinear, hence yields a map $g' : P \otimes R \rightarrow (P/Q) \otimes R$, which further factors through the quotient by $Q \otimes R$, yielding a map $g : (P \otimes R)/(Q \otimes R) \rightarrow (P/Q) \otimes R$. And f, g are mutually inverse, yielding the claim. $\square$

By the claim, g_* induces an isomorphism (since functors send isos to isos)

$$M'' \otimes N \leftarrow (M'/\text{im}(f)) \otimes N \cong (M' \otimes N)/\text{im}(f_*)$$

so $\text{im}(f_*) = \ker(g_*)$. And surjectivity of g_* follows directly from surjectivity of g . $\square$